

GDP vs NEET Analysis

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2025-12-03

Import Data

```
gdp_df <- read_csv("/Users/vasilikimichalia/Desktop/IDS_Group_Proj-main/Exported_data/gdp_per_capita_df")

## Rows: 1176 Columns: 9
## -- Column specification -----
## Delimiter: ","
## chr (3): Entity, Code, Continent
## dbl (5): Year, Gdp_per_capita, Growth, Population, Gdp
## lgl (1): Ldc
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

neet_df <- read_csv("/Users/vasilikimichalia/Desktop/IDS_Group_Proj-main/Exported_data/youth_not_in_eet")

## Rows: 615 Columns: 7
## -- Column specification -----
## Delimiter: ","
## chr (3): Entity, Code, Continent
## dbl (4): Year, Perc_youth_not_in_eet, Perc_change, Population
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Merge & Clean

```
# Perform an inner join to combine GDP and NEET datasets
# Matching rows where Entity (country name), Code (country code), and Year all align
merged_df <- inner_join(gdp_df, neet_df, by = c("Entity", "Code", "Year"))

# Clean and restructure the merged dataset
merged_df <- merged_df %>%
  # Rename columns for clarity (Entity -> Country, keep only one Continent column)
  rename(
    Country = Entity,
    Continent = Continent.x
  ) %>%
  # Select only the columns needed for analysis
  select(Continent, Country, Year, Gdp_per_capita, Perc_youth_not_in_eet) %>%
  # Convert GDP and NEET columns to numeric type to ensure proper calculations
```

```
mutate(
  Gdp_per_capita      = as.numeric(Gdp_per_capita),
  Perc_youth_not_in_eet = as.numeric(Perc_youth_not_in_eet)
) %>%
# Remove any rows with missing values to ensure complete cases for analysis
na.omit()

# Convert Continent to a factor (categorical variable) for grouping and visualization
merged_df$Continent <- as.factor(merged_df$Continent)
```

Global Correlation

```
# Perform a Pearson correlation test at the global (country) level
# Tests the linear relationship between GDP per capita and NEET percentage
# across all countries and years in the dataset
global_cor <- cor.test(
  merged_df$Gdp_per_capita,
  merged_df$Perc_youth_not_in_eet,
  method = "pearson"
)

# Display correlation results (r value, p-value, confidence interval)
global_cor
```

```
##
## Pearson's product-moment correlation
##
## data: merged_df$Gdp_per_capita and merged_df$Perc_youth_not_in_eet
## t = -22.296, df = 609, p-value < 2.2e-16
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.7118572 -0.6242660
## sample estimates:
## cor
## -0.6703905
```

Global Plot

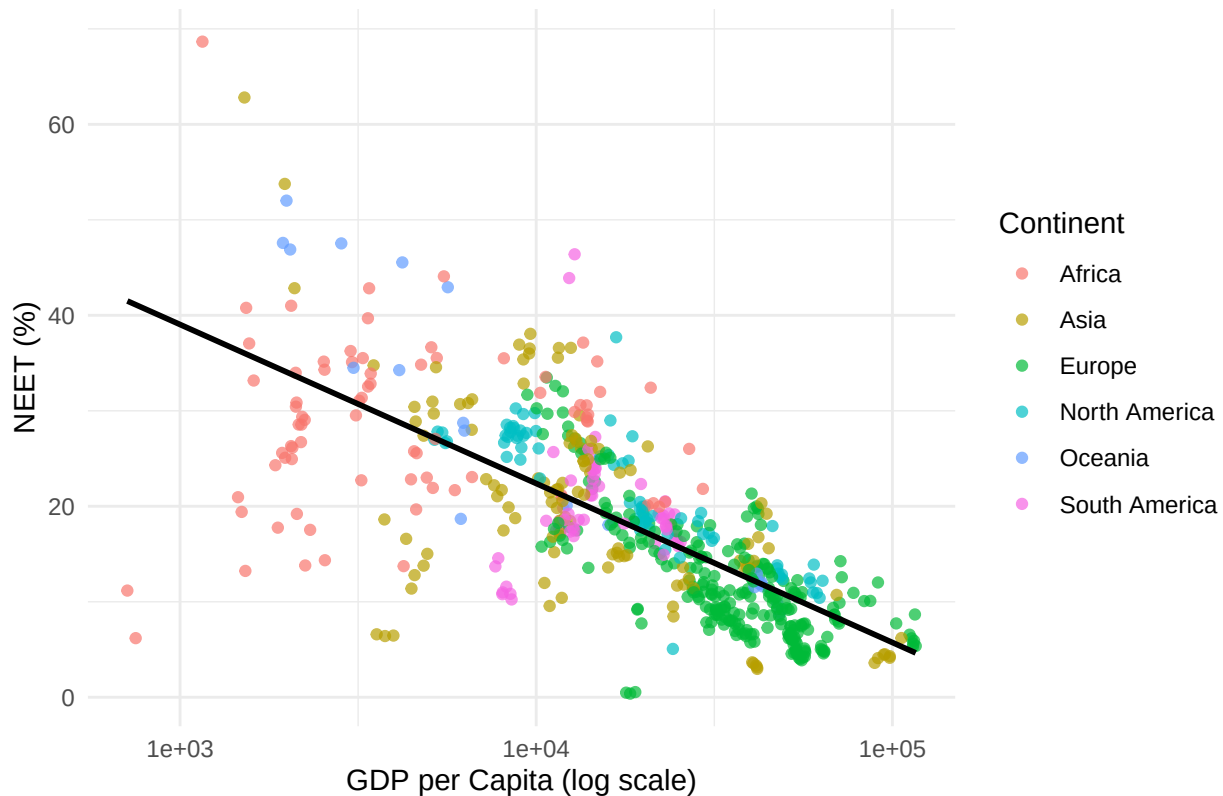
```
# Create a scatter plot showing the relationship between GDP and NEET at country level
g1 <- ggplot(merged_df, aes(Gdp_per_capita, Perc_youth_not_in_eet, color = Continent)) +
# Add semi-transparent points, colored by continent
geom_point(alpha = 0.7) +
# Add a black linear regression line (without confidence interval shading)
geom_smooth(method = "lm", se = FALSE, color = "black") +
# Use log scale on x-axis to better visualize the wide range of GDP values
scale_x_log10() +
# Add informative labels
labs(
  title = "GDP per Capita vs NEET (%) - Global",
  x = "GDP per Capita (log scale)",
  y = "NEET (%)"
```

```
) +
# Apply a clean, minimal theme
theme_minimal()

# Display the plot
g1
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

GDP per Capita vs NEET (%) – Global



Continent Means + Correlation

```
# Aggregate data to the continent level by calculating mean values
continent_means <- merged_df %>%
  # Group all observations by continent
  group_by(Continent) %>%
  # Calculate the mean GDP and mean NEET percentage for each continent
  summarise(
    Mean_GDP = mean(Gdp_per_capita),
    Mean_NEET = mean(Perc_youth_not_in_eet),
    .groups = "drop" # Remove grouping structure after summarizing
  )

# Perform correlation test on continent-level aggregated data
# This tests whether continents with higher average GDP have lower average NEET
continent_cor <- cor.test(
  continent_means$Mean_GDP,
```

```

continent_means$Mean_NEET,
method = "pearson"
)

# Display correlation results for continent-level analysis
continent_cor

##
## Pearson's product-moment correlation
##
## data: continent_means$Mean_GDP and continent_means$Mean_NEET
## t = -3.4708, df = 4, p-value = 0.02557
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
## -0.9852232 -0.1848900
## sample estimates:
## cor
## -0.866442

```

Continent Plot

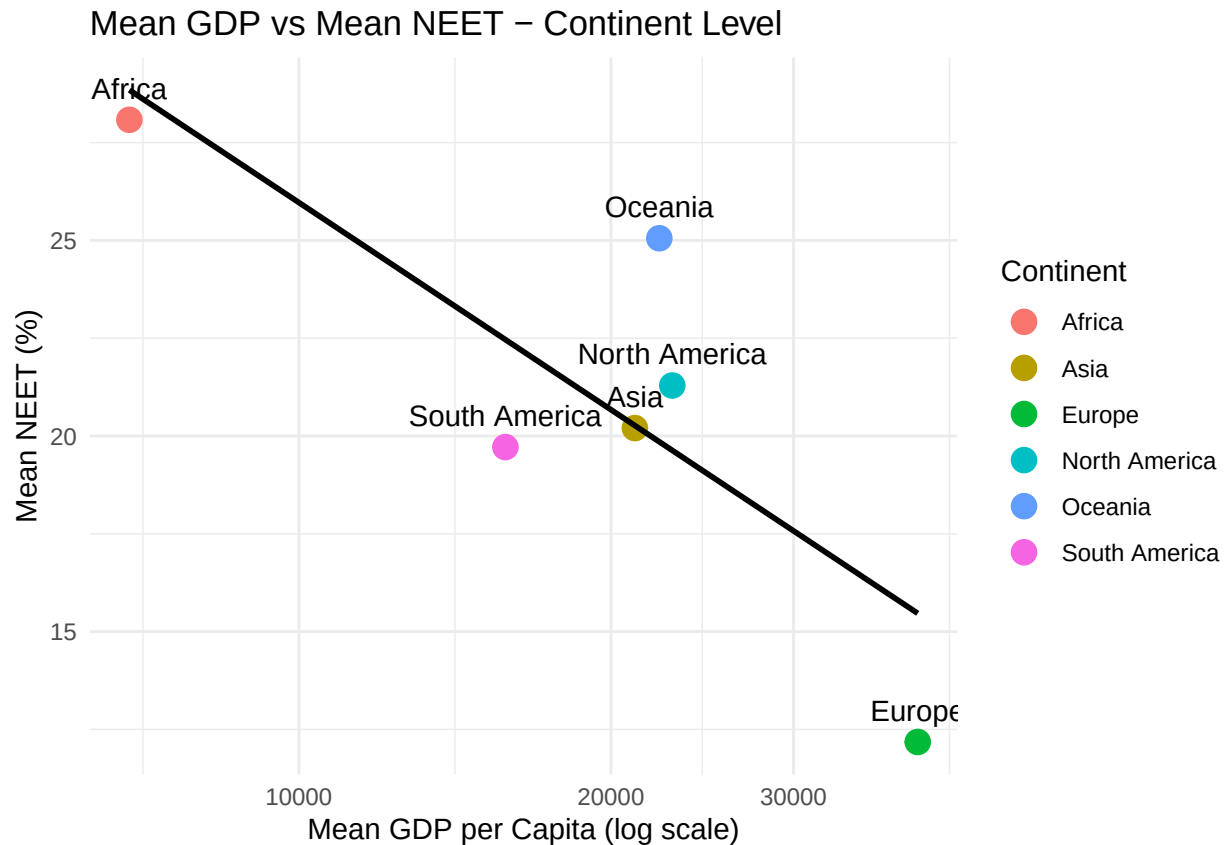
```

# Create a scatter plot of the 6 continent means
g2 <- ggplot(continent_means, aes(Mean_GDP, Mean_NEET)) +
  # Add larger points for each continent, colored by continent
  geom_point(size = 4, aes(color = Continent)) +
  # Label each point with the continent name, positioned above the point
  geom_text(aes(label = Continent), vjust = -1) +
  # Add a black linear regression line showing the continent-level trend
  geom_smooth(method = "lm", se = FALSE, color = "black") +
  # Use log scale on x-axis for consistency with global plot
  scale_x_log10() +
  # Add informative labels
  labs(
    title = "Mean GDP vs Mean NEET - Continent Level",
    x = "Mean GDP per Capita (log scale)",
    y = "Mean NEET (%)"
  ) +
  # Apply clean, minimal theme
  theme_minimal()

# Display the continent-level plot
g2

## `geom_smooth()` using formula = 'y ~ x'

```



```
# Create a summary table comparing the two levels of analysis
# Build a clean hypothesis test results table
results_table <- data.frame(
  # Label each analysis level
  Level = c("Global (Countries)", "Continent Means (6 points)"),
  # Extract Pearson correlation coefficients from both tests
  Pearson_r = c(global_cor$estimate, continent_cor$estimate),
  # Extract p-values to assess statistical significance
  p_value = c(global_cor$p.value, continent_cor$p.value),
  # Record sample sizes (number of observations in each analysis)
  N = c(nrow(merged_df), nrow(continent_means)),
  # Determine if results are statistically significant at the alpha level
  Significant = c(global_cor$p.value < ALPHA,
                  continent_cor$p.value < ALPHA)
)

# Display the results table for easy comparison
results_table
```

```
##           Level  Pearson_r    p_value  N Significant
## 1   Global (Countries) -0.6703905 5.771296e-81 611      TRUE
## 2 Continent Means (6 points) -0.8664420 2.556541e-02   6      TRUE
```